Yuetang Liang

Preferred Name: Michelle

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Objective

I am a Honours Computer Science (Co-op) student at the University of Ottawa with a strong foundation in C/C++, systems programming, and Linux-based development. I have experience developing automated testing tools, executing functional validation workflows, and debugging system-level issues in controlled lab environments. I am particularly interested in hardware-software interaction, system validation, and verification methodologies for complex computing systems. I bring strong analytical skills, attention to detail, and a structured approach to testing and problem-solving.

Education

University of Ottawa – Ottawa, ON
Honours Computer Science (Co-op) Undergraduate Program
Expected Graduation: around 2028

Relevant Courses:

- Systems Programming & Algorithms
- Data Structures and Software Design
- Digital Design Verilog
- Linear Algebra
- Probability & Statistics for Computing
- Java
- Python
- C++

Academic Achievement:

- Completed 1000+ LeetCode problems with focus on algorithms, debugging, edge-case analysis, and code optimization.
- Demonstrated strong problem-solving ability through medium to advanced-level algorithm challenges.

Achieved proficiency in solving **medium to hard-level coding challenges**, demonstrating analytical thinking and systematic problem-solving skills.

Technical Skills

- C++ (object-oriented design, data structures, debugging)
- Python (automation, validation scripts, log analysis)
- Bash (Linux scripting and workflow automation)
- Java
- C/C++ and OOP
- Linux / Unix
- Test development and validation
- Network protocols (L2/L3, TCP/IP, UDP, Ethernet)
- Simulator / tool development mindset
- Automation (Python, MATLAB-like workflows)
- Verification and quality focus

Systems & Development

- Unix/Linux environment (process control, scripting, system debugging)
- Git version control
- Virtualized environments (VMware, VirtualBox)

Testing & Validation

- Functional test development and execution
- Automated test scripting
- Log analysis and debugging
- Structured test planning and repeatable validation workflows

Networking & Protocols

- TCP/IP, UDP, Ethernet (L2/L3 fundamentals)

- Packet analysis using Wireshark
- Service scanning and validation (Nmap)

Experience

Systems Validation & Performance Research Assistant – University of Ottawa, Ottawa, ON
[September 2025 – December 2025]

In this role, I contributed to the structured validation and performance analysis of computing systems, with a focus on simulation modeling, Linux-based system behavior, and debugging.

- I developed automation tools in Python and C++ to simulate system behavior under varying workloads, enabling repeatable testing and analysis of runtime characteristics.
- I designed and executed structured validation workflows to evaluate process scheduling, CPU utilization, and memory access patterns.
- I conducted performance analysis by tracing execution paths, identifying bottlenecks related to compute and memory hierarchy, and suggesting optimizations.
- I created lightweight simulation environments that mimicked real-world system loads, providing insight into hardware-software interaction under stress.
- I applied systematic debugging techniques to resolve configuration issues and runtime anomalies, ensuring accurate validation results.
- I documented testing procedures, simulation designs, and performance findings to ensure clarity, reproducibility, and effective knowledge sharing within the research team.

This role strengthened my understanding of performance-driven system evaluation, memory and compute analysis, and structured simulation modeling.

Systems Engineering & Linux Performance Debugging Project – University of Ottawa, Ottawa, ON
[September 2025 – December 2025]

I supported system stability and performance optimization by developing automation tools, analyzing runtime logs, and diagnosing system-level inefficiencies in a Linux environment.

- I created Python and Bash scripts to automate system monitoring, collect performance metrics, and generate validation reports.
- I performed log-based analysis to trace the root cause of performance issues related to process scheduling, system calls, and resource contention.
- I worked with virtualized environments to model system configurations and validate behavioral consistency across different deployment scenarios.

- I conducted packet-level inspection using Wireshark to validate TCP/IP, UDP, and Ethernet communication protocols and ensure correct system interactions.
- I collaborated with academic teams to ensure structured, reproducible validation workflows, contributing to a high level of system reliability and test coverage.

This experience deepened my skills in debugging complex systems, interpreting runtime behavior, and working across software and system layers.

Academic Project: Simulation Framework for Networked Systems – University of Ottawa
[September 2025]

In this academic project, I developed a modular simulation and analysis framework to evaluate system performance in a virtualized networking environment.

- I implemented object-oriented components in Python and C++ to simulate communication between nodes and measure system throughput and latency.
- I developed structured test cases to validate correctness, protocol adherence, and communication efficiency under varied conditions.
- I applied performance metrics to evaluate the behavior of simulated systems and identify areas for optimization.
- I documented the system architecture, testing procedures, and performance results to support reproducibility and transparency in analysis.

This project allowed me to strengthen my skills in simulation modeling, system analysis, and structured testing.

Certifications

- Ready for Cisco Certified Network Associate (CCNA) (in progress)

Core Competencies

I have a strong foundation in logical reasoning and problem-solving, combined with programming and mathematical skills. I am adaptable, eager to learn, and capable of approaching complex technical challenges methodically. I can work effectively in teams, communicate technical concepts clearly, and apply my knowledge to analyze and improve system security.

Relevant Coursework & Academic Achievement

ITI1100 – Digital Systems I
I developed a strong foundation in digital logic and computer hardware fundamentals, including

binary representation, logic gates, Boolean algebra, combinational and sequential circuits, and basic processor organization. I learned how software interacts with hardware at the logic level, which strengthened my understanding of system behavior and hardware-software integration.

ITI1121 – Introduction to Computing II (C++/Java)
I applied object-oriented programming principles using C++/Java, including classes, encapsulation, inheritance, and modular design. I developed structured programs, implemented algorithms, and practiced debugging and code optimization. This course strengthened my ability to write maintainable, efficient, and scalable software.

MAT1322 – Calculus II
I developed strong analytical and problem-solving skills through advanced calculus topics including integration techniques, series, and mathematical modeling. I applied mathematical reasoning to analyze system behavior and solve complex multi-step problems relevant to computing and engineering.

MAT1348 – Discrete Mathematics for Computing
I built a foundation in discrete structures essential for computer science, including logic, proofs, set theory, combinatorics, graph theory, and recursion. This course strengthened my ability to think logically, analyze algorithms, and design efficient computational solutions.

ITI1120 – Introduction to Computing I
I have achieved strong results by mastering fundamental programming concepts, including algorithm design, data structures, recursion, and problem decomposition. Demonstrated the ability to write clean, well-structured Python code and apply logical thinking to solve increasingly complex computational problems.

MAT1341 – Introduction to Linear Algebra
I am excelled in vector spaces, matrices, Gaussian elimination, orthogonality, projections, and least-squares approximation. Applied rigorous mathematical reasoning to solve multi-step algebraic problems and connect linear algebra concepts to computer science applications such as graphics, optimization, and data analysis.

MAT1320 – Calculus I
I have earned high academic standing through strong conceptual understanding of limits, derivatives, optimization, and mathematical modeling. Applied systematic problem-solving techniques and demonstrated precision in symbolic manipulation, logical reasoning, and multi-stage calculations.

The following results for reference:

MAT1341 E00 Intro. To Linear Algebra [LEC] 20259

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Grades

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Grade Item	Points	Weight Achieved	Grade	Comments and Assessments
Midterm 1	37 / 40	13.88 / 15	A+	
Midterm 2	38 / 40	14.25 / 15	A+	
Midterm 3	34 / 40	12.75 / 15	A	

MAT1320 B00 Calculus I [LEC] 20259

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Grades

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Grade Item	Points	Weight Achieved	Grade	Comments and Assessments
Getting to know Möbius	- / 5	- / 0	-	
Assignment 1	10 / 10	0 / 0	A+	
Assignment 2	10 / 10	0 / 0	A+	

ITI1120 C04 Introduction To Computing I 20259

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Grades

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Grade Item	Points	Weight Achieved	Grade	Comments and Assessments
Quizzes		5 / 5	A+	
Quiz 1	5 / 5	2.5 / 2.5	A+	
Quiz 2	22 / 22	2.5 / 2.5	A+	View Quiz Attempts
Assignments		18 / 18	A+	
Assignment 1	80 / 80	9 / 9	A+	
Assginment 2	100 / 100	9 / 9	A+	
Assginment 3	- / 100	- / -	-	
Assginment 4	- / 100	- / -	-	
Midterm	23 / 25	0 / 0	A+	
Midterm adjusted	25 / 25	25 / 25	A+	